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Vehicle front body structure

Abstract

A vehicle front body structure includes a duct member that surrounds a radiator on an inner side in a vehicle width direction of a right and left pair of crash cans. The duct member is a member that completely surrounds the radiator and includes side duct sections and an upper duct section. The side duct sections and the upper duct section are connected such that, when the duct member receives a specified load from a bumper member of a body during a frontal collision, the side duct sections and the upper duct section are disengaged from each other in the engagement sections.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Japanese Patent Application 2021-087814, filed May 25, 2021, the entire contents of which are incorporated herein by reference.

BACKGROUND

Field

(2) The present disclosure relates to a vehicle front body structure and, in particular, to a vehicle front body structure including: a right and left pair of crash cans; a radiator provided on an inner side in a vehicle width direction of these crash cans; and a duct member that surrounds the radiator on the inner side in the vehicle width direction of the crash cans.

Description of the Related Art

(3) Conventionally, in order to introduce air into a heat exchanger (a radiator or the like), which is provided in an engine compartment of a vehicle, to improve cooling capacity, such a structure has been known that is provided with a duct member for surrounding the heat exchanger and extending forward, so as to completely surround the heat exchanger (for example, Patent document 1).

PRIOR ART DOCUMENTS

Patent Documents

(4) [Patent document 1] JP-A-2007-91061

SUMMARY

Technical Problems

(5) Here, in the case where the duct member that completely surrounds the heat exchanger is provided as in Patent document 1, the air that has been introduced into a front bumper can flow linearly toward the heat exchanger. In this way, the cooling capacity of the heat exchanger can be improved. Furthermore, since the air flows linearly in the front bumper, aerodynamic drag of the vehicle can also be improved. For such a reason, it is useful to provide the duct member in the front bumper so as to rectify the air.

(6) In order to cause the air to flow linearly toward the heat exchanger, such a duct structure that completely surrounds the heat exchanger as in Patent document 1 is preferably adopted. However, in the case where the duct member as a highly-rigid integral structure that completely surrounds the heat exchanger is provided, it is concerned that the duct member hinders deformation of a crash can during a frontal collision.

(7) Due to such concern, as a rectification structure in the front bumper, a plane member that partially surrounds the heat exchanger for rectification is generally provided to ensure collision performance, instead of adopting the highly-rigid duct member that completely surrounds the heat exchanger as described above.

(8) Meanwhile, in order to improve cooling performance of the heat exchanger, the present inventors have keenly studied a body structure capable of satisfying the requested collision performance while adopting the duct structure that completely surrounds the heat exchanger.

(9) In view of the above, the present disclosure has been made to solve the above-described problem and therefore has a purpose of providing a vehicle front body structure capable of simultaneously improving cooling performance by a heat exchanger and exerting collision performance of a vehicle.

Means for Solving the Problems

(10) In order to solve the above-described problem, the present disclosure is a vehicle front body structure including: a right and left pair of crash cans, each of which extends in a vehicle longitudinal direction; a radiator that is provided on an inner side in a vehicle width direction of the right and left pair of these crash cans; and a duct member that surrounds the radiator on the inner side in the vehicle width direction of the right and left pair of crash cans. The duct member is a member that completely surrounds the radiator and has a vertical wall section and a transverse wall section. The vertical wall section and the transverse wall section of the duct member have at least one engagement section in which those vertical wall section and transverse wall section are engaged with and connected to each other. The vertical wall section and the transverse wall section of the duct member are connected such that, when the duct member receives a specified load from a bumper member of a body during a frontal collision, the vertical wall section and the transverse wall section are disengaged from each other in the at least one engagement section.

(11) According to the present disclosure that is configured as described above, since the duct member completely surrounds the radiator, it is possible to ensure an air flow toward a heat exchanger. In addition, when a collision load is input from the front, the vertical wall section and the transverse wall section of the duct member are disengaged from each other in the at least one engagement section. Thus, it is possible to reduce strength of the duct member against the load

from the front of the vehicle, and thereby deformation of the crash can is not hindered during the frontal collision. As the result of these, the present disclosure can simultaneously improve cooling performance by the heat exchanger and exert collision performance of the vehicle.

(12) In the present disclosure, preferably, the transverse wall section of the duct member is formed with at least one notch that is opened rearward, and the at least one engagement section is configured by engaging a part of the vertical wall section with the notch of the transverse wall section.

(13) According to the present disclosure that is configured as described above, the part of the vertical wall section is engaged with the notch, which is opened rearward, in the transverse wall section in the at least one engagement section. Thus, when the vertical wall section receives the collision load from the front, the vertical wall section is displaced rearward in the vehicle relative to the transverse wall section, and is disengaged from the notch of the transverse wall section. In this way, it is possible to further effectively reduce the strength of the duct member against the load from the front of the vehicle.

(14) In the present disclosure, preferably, the vertical wall section of the duct member has a frame section and an inner surface section having lower strength than this frame section. In the at least one engagement section, a part of the frame section of the vertical wall section is connected to the transverse wall section in a manner to be disengaged therefrom.

(15) According to the present disclosure that is configured as described above, when the vertical wall section receives the collision load from the front, the frame section, which is a part of a side of the duct member, in the vertical wall section is deformed. In this way, is possible to effectively reduce the strength of the duct member against the load from the front of the vehicle. In particular, since the inner surface section has the lower strength than the frame section, it is possible to suppress the inner surface section from hindering the deformation of the frame section.

(16) In the present disclosure, preferably, the vertical wall section of the duct member is formed with a recessed section, and the recessed section is formed with a ridgeline that extends outward in the vehicle width direction.

(17) According to the present disclosure that is configured as described above, when the vertical wall section receives the collision load from the front during the frontal collision, such a load is transmitted outward in the vehicle width direction along the ridgeline of the recessed section. In this way, the vertical wall section can be deformed outward in the vehicle width direction to have a V-shape. Along with such deformation, the vertical wall section and the transverse wall section can further effectively be disengaged from each other in the at least one engagement section.

(18) In the present disclosure, preferably, the bumper member of the body includes a bumper reinforcement that extends in the vehicle width direction on a front surface of a vehicle, and in a vehicle vertical direction of the vertical wall section, the recessed section of the vertical wall section in the duct member is formed at substantially the same height position as the bumper reinforcement.

(19) According to the present disclosure that is configured as described above, it is possible to effectively transmit the collision load, which is input from the bumper reinforcement, to the recessed section of the vertical wall section. In this way, the vertical wall section can further reliably be deformed outward in the vehicle width direction to have the V-shape.

Advantages

(20) The vehicle front body structure of the present disclosure can simultaneously improve the cooling performance by the heat exchanger and exert the collision performance of the vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view in which a front portion of a vehicle having a vehicle front body structure according to an embodiment of the present disclosure is seen obliquely downward from a vehicle right side.
- (2) FIG. 2 is a perspective view in which a radiator, a radiator duct, and a body shock absorbing member provided in the vehicle front body structure according to this embodiment are seen obliquely downward from the vehicle right side.
- (3) FIG. 3 is a perspective view illustrating the radiator illustrated in FIG. 2 and a duct member.
- (4) FIG. 4 is a partial cross-sectional side view illustrating the radiator and the duct member illustrated in FIG. 3.
- (5) FIG. 5 is a partial enlarged perspective view enlarging an engagement section between a side duct section and an upper duct section of the duct member illustrated in FIG. 3.
- (6) FIG. 6 is a partial enlarged perspective view enlarging a recessed section that is formed in the side duct section of the duct member illustrated in FIG. 3.
- (7) FIG. 7 is a partial enlarged plan view in which the side duct section and the recessed section illustrated in FIG. 6 are seen from above.
- (8) FIG. 8 is a partial enlarged perspective view enlarging a locking section of the radiator duct to the shock absorbing member according to this embodiment.
- (9) FIGS. 9A and 9B include partial enlarged perspective views, each of which illustrates an exemplary deformation state of the duct member caused by load input from front of the vehicle to the vehicle having the vehicle front body structure according to this embodiment in an early phase of a collision.
- (10) FIG. 10 is a partial enlarged perspective view for illustrating a deformation state of the side duct section at the time when the side duct section and the upper duct section of the duct member according to this embodiment are disengaged from each other.

DESCRIPTION OF EMBODIMENTS

- (11) A description will hereinafter be made on a vehicle front body structure according to an embodiment of the present disclosure with reference to the accompanying drawings.
- (12) First, a description will be made on the vehicle front body structure according to the embodiment of the present disclosure with reference to FIG. 1 and FIG. 2. FIG. 1 is a perspective view in which a front portion of a vehicle having the vehicle front body structure according to the embodiment of the present disclosure is seen obliquely downward from a vehicle right side. FIG. 2 is a perspective view in which the vehicle front body structure according to this embodiment is seen obliquely downward from the vehicle right side.
- (13) First, as illustrated in FIG. 1, a vehicle 1 having the vehicle front body structure according to the embodiment of the present disclosure includes, in a front portion thereof: a hood 4 that covers an engine compartment 2; an upper grille panel 6 that extends in a vehicle width direction on a front side of the hood 4; a bumper fascia 8 that is provided below this upper grille panel 6 and on a front surface of the vehicle 1; a front grille 10 that is provided between the upper grille panel 6 and the bumper fascia 8; a headlamp unit 12 that is arranged on each of right and left sides of this front grille 10 and is arranged in a portion above the bumper fascia 8; and a front fender 14 that is provided on each of right and left sides of the hood 4 and is arranged on a lateral surface of the vehicle.
- (14) Next, as illustrated in FIG. 2, the vehicle 1 includes, as frame members constituting a body: a right and left pair of front side frames 16, each of which extends in a vehicle longitudinal direction of the engine compartment 2; a crash can 18 as a shock absorbing member that is provided to a front end portion of each of these front side frames 16; and a bumper reinforcement 20 that is attached to front end portions of these crash cans 18 and extends in the vehicle width direction to receive a load during a frontal collision.

(15) In addition, in the front portion of the vehicle **1**, a right and left pair of apron members is provided on an upper side and both of right and left sides of the engine compartment **2**, and extends in the vehicle longitudinal direction. Each of the apron members has a tip portion that is curved inward in the vehicle width direction, and such tip portions are connected to each other via an upper shroud member.

(16) At a position below the upper shroud member, a radiator **22** is provided to cool an engine with air that is introduced therein through the front grille **10**. This radiator **22** is provided with: a grille shutter **24** that is provided on a front surface thereof; and a radiator shroud **26** that extends in a rectangular shape so as to cover a radiator core **36**, which will be described below and illustrated in FIG. **4**, a cooling fan **38** of the radiator core **36**, and the like.

(17) A duct member **28** that extends in a rectangular shape in front of the radiator **22** is connected to a front end portion of this radiator shroud **26**. As will be described below, the radiator **22** is completely surrounded by the duct member **28** along with the radiator shroud **26**. The radiator **22**, the radiator shroud **26**, and the duct member **28** are provided on an inner side in the vehicle width direction of the right and left crash cans **18**. Here, the radiator shroud **26** and the duct member **28** may integrally be molded to have such a configuration that does not hinder deformation of the duct member **28**, which will be described below.

(18) Next, the vehicle **1** includes a grille reinforcement **30** (denoted by a reference sign **30** in FIGS. **9A** and **9B** and FIG. **10**). The grille reinforcement **30** is attached to an unillustrated support bracket on the body side, and the like, and mainly supports and reinforces the bumper fascia **8** and the front grille **10**. This grille reinforcement **30** is a member that is provided on an inner side of the bumper fascia **8** and the front grille **10** illustrated in FIG. **1** and extends along the bumper fascia **8** and the front grille **10**. In this embodiment, this grille reinforcement **30** and the bumper reinforcement **20** primarily function as bumper members that receive a shock from an obstacle (another vehicle, an upper limb portion of a pedestrian, or the like) in front of the vehicle.

(19) Next, a pair of load transmission members **32** is provided on a rear side of the grille reinforcement **30** and a front side of the bumper reinforcement **20**, and extends vertically in a manner not to hinder an air flow into the radiator **22**. As will be described below, each of these load transmission member **32** is provided as an abutment member that abuts the duct member **28** in an early phase of the collision, and plays a role of transmitting the load to the duct member **28** when the collision load is applied to the grille reinforcement **30**.

(20) A seal member **34** is attached to each of the load transmission members **32**, and has a function of sealing the respective load transmission member **32** and the grille reinforcement **30**. The load transmission members **32** are fixed to the grille reinforcement **30** via these seal members **34**. As will be described below, each of these seal members **34** also plays the role of transmitting the load to the duct member **28** in the early phase of the collision. The seal members **34** and the load transmission members **32** are attached to such a grille reinforcement **30**.

(21) Furthermore, as will be described below, the bumper reinforcement **20** also plays the role of transmitting the load to the duct member **28** in the early phase of the collision.

(22) Next, a description will be made on configurations of the radiator **22** and the duct member **28** with reference to FIG. **3** and FIG. **4**. FIG. **3** is a perspective view illustrating the radiator and the duct member illustrated in FIG. **2**. FIG. **4** is a partial cross-sectional side view illustrating the radiator and the duct member illustrated in FIG. **3**. FIG. **4** is a view illustrating cross sections of central portions in the vehicle width direction of the radiator and the duct member illustrated in FIG. **3**, and is a view of a state where an inner surface of a left side duct section of the duct member is visible.

(23) First, as illustrated in FIG. **4**, the radiator **22** includes the grille shutter **24**, the radiator core **36**, the cooling fan **38**, and the like. As illustrated in FIG. **3** and FIG. **4**, the radiator shroud **26** accommodates this radiator **22**.

(24) Together with the radiator shroud **26**, the duct member **28** is provided to completely surround

the radiator **22**. In this embodiment, the duct member **28** extends to completely surround the radiator **22**, so as to completely cover at least the grille shutter **24** of the radiator **22**.

(25) Next, as illustrated in FIG. **3**, the duct member **28** has: an upper duct section (an upper transverse wall section) **40** on an upper surface thereof; a lower duct section (a lower transverse wall section) **42** on a lower surface thereof; and side duct sections (vertical wall sections) **44**, **46** on both of lateral surfaces in the vehicle width direction thereof. The side duct sections **44**, **46** extend in a vehicle vertical direction in a manner to connect both of edge portions (both of right and left edge portions) in the vehicle width direction of the upper duct section **40** and the lower duct section **42**.

(26) First, the upper duct section **40** is configured as a resin member that continuously extends from a front end portion of an upper surface of the radiator shroud **26**. The lower duct section **42** is configured as a resin member that continuously extends from a front end portion of a lower surface of the radiator shroud **26**. Members constituting this lower duct section **42** include lower side sections **48** that extend upward in the vehicle from both of edge portions in the vehicle width direction of the lower duct section **42**. These lower side sections **48** constitute parts near lower edge portions of the side duct sections (the vertical wall sections) **44**, **46**.

(27) Next, as illustrated in FIG. **3**, each of the side duct sections **44**, **46** has the above-described lower side section **48** and an upper side section **50** on top of the lower side section **48**.

(28) First, each of the right and left lower side sections **48** has a frame section **48a** that is made of a resin, continues from respective one of the right and left edge portions of the above-described lower duct section **42**, and extends along an outer edge portion of the lower side section **48**. An inner surface section **48b** of each of these frame sections **48a** is constructed of a plane member that is made of rubber and has lower strength than the resin frame section **48a**. This rubber-made inner surfaces **48b** is attached to the frame section **48a** by clipping a peripheral edge portion of the inner surfaces **48b** to the frame section **48a** (also see an enlarged view in FIG. **6**).

(29) Next, as illustrated in FIG. **3**, each of the right and left upper side sections **50** has a frame section **50a** that is made of the resin and extends along an outer peripheral edge of the respective upper side section **50**. Each of these resin frame sections **50a** is formed of a different member from the resin member that constitutes the upper duct section **40**. However, the upper duct section **40** and the frame section **50a** of the upper side section **50** may be formed of an integrally-molded resin member. An inner surface **50b** of each of these frame sections **50a** is constructed of a plane member that is made of the rubber and has lower strength than the respective resin frame section **50a**. This rubber-made plane member **50b** is attached to the frame section **50a** by clipping a peripheral edge portion of the plane member **50b** to the frame section **50a** (also see an enlarged view in FIG. **5**).

(30) Here, as illustrated in FIG. **2**, the load transmission members **32** and the seal members **34** described above are provided at positions that are located below the upper duct section **40** and above the lower duct section **42** and oppose front end portions of the side duct sections **44**, **46**. In this way, as will be described below, when the load is transmitted from the grille reinforcement **30** in the early phase of the collision, the load transmission members **32** and the seal members **34** transmit the load to the side duct sections **44**, **46**, and thereby promote the deformation of the duct member **28**.

(31) Next, a description will be made on a coupling structure (a first deformation promoting structure) between the upper duct section **40** and each of the side duct sections **44**, **46** and in the duct member **28** with reference to FIG. **3** and FIG. **5**. FIG. **5** is a partial enlarged perspective view enlarging an engagement section between the side duct section and the upper duct section of the duct member illustrated in FIG. **3**.

(32) As illustrated in FIG. **3** and FIG. **5**, a notch **52** is formed in each of right and left edge portions of the upper duct section **40**, and is opened rearward in the vehicle.

(33) An engagement vertical wall portion **50c**, as a part of the resin member constituting the upper side section **50**, is fitted into a front portion **52a** of each of these notches **52**. This front portion **52a**

extends in the vehicle longitudinal direction with a constant width such that the upper side section **50** is engaged with the upper duct section **40** at this position until the load in a specified amount or larger is applied.

(34) Meanwhile, a rear portion **52b** of the notch **52** is notched obliquely outward in the vehicle width direction so as to guide rearward movement and deformation of the upper side section **50** at the time when respective one of the side duct sections **44**, **46** moves rearward with respect to the upper duct section **40** as will be described below.

(35) Here, the notch **52** and the engagement vertical wall portion **50c**, as the part of the upper side section **50** engaged with the notch **52**, constitute an engagement section **54** between the upper duct section (the transverse wall section) **40** and respective one of the side duct sections (the vertical wall sections) **44**, **46**. In detail, the notch **52** and the engagement vertical wall portion **50c** constitute the engagement section **54** between the upper duct section **40** and the upper side section **50**.

(36) In this embodiment, as will be described below, in the early phase of the collision, the upper duct section (the transverse wall section) **40** and each of the side duct sections (the vertical wall sections) **44**, **46** are disengaged from each other in such an engagement section **54**, and the deformation of the duct member **28** is thereby promoted.

(37) In this embodiment, the engagement sections **54** are provided at two right and left positions above the duct member **28**. However, as a modified example, one of the engagement sections **54** may be formed in the same manner, and the other thereof may be formed to have a fixing structure that hinders the load transmission during the collision. Alternatively, the engagement sections **54** may be provided below the duct member **28**.

(38) Next, a description will be made on a structure (a second deformation promoting structure) of the lower side section **48** in each of the side duct sections **44**, **46** with reference to FIG. 3, FIG. 6, and FIG. 7. FIG. 6 is a partial enlarged perspective view enlarging a recessed section that is formed in the side duct section of the duct member illustrated in FIG. 3. FIG. 7 is a partial enlarged plan view in which the side duct section and the recessed section illustrated in FIG. 6 are seen from above.

(39) First, as illustrated in FIG. 3, FIG. 6, and FIG. 7 in each of the side duct sections **44**, **46**, the resin frame section **48a** of the lower side section **48** is formed with a recessed section **56** at a boundary position thereof with the upper side section **50**. The recessed section **56** is a concave section that extends downward in the vehicle and outward in the vehicle width direction in an arc shape. As illustrated in FIG. 6 and FIG. 7, each of these recessed sections **56** is opened to the inner side in the vehicle width direction. The frame section **48a** is also formed with a ridgeline **56a** by the respective recessed section **56** that extends outward in the vehicle width direction.

(40) Here, as illustrated in FIG. 6, the frame section **50a** and the rubber member **50b**, which is the inner surface of the frame section **50a**, in the upper side section **50** extend on the outer side in the vehicle width direction of the recessed section **56**.

(41) In addition, as illustrated in FIG. 7, the frame section **50a** of this upper side section **50** extends to have an L-shaped cross section on the outer side in the vehicle width direction of the recessed section **56** and a vehicle rear side thereof in a plan view. A portion on the vehicle rear side of the frame section **50a** is clipped to a front surface of the radiator shroud **26**. Meanwhile, the rubber member **50b** as the inner surface extends to have an L-shape on the outer side in the vehicle width direction of the recessed section **56** and a vehicle front side thereof in the plan view. A portion on the outer side in the vehicle width direction of the rubber member **50b** is clipped to the frame section **50a**.

(42) Next, a description will be made on attachment and arrangement of the side duct sections **44**, **46** to the bumper reinforcement **20** with reference to FIG. 7 and FIG. 8. FIG. 8 is a partial enlarged perspective view enlarging a locking section of a radiator duct to the shock absorbing member according to this embodiment.

(43) First, as illustrated in FIG. 8, the rubber member **50b** of each of the side duct sections **44, 46** is partially locked to the bumper reinforcement **20**.

(44) In addition, as illustrated in FIG. 8, the recessed section **56** of the lower side section **48** (FIG. 8 illustrates a portion on the outer side in the vehicle width direction of the recessed section **56**) is formed at substantially the same height position as the bumper reinforcement **20** in the vehicle vertical direction of each of the side duct sections **44, 46**. In this way, when the bumper reinforcement **20** abuts the side duct sections **44, 46** in the early phase of the collision, the load is easily transmitted to these recessed sections **56**.

(45) In this early phase of the collision, as indicated by a reference sign A in FIG. 7, a load vector in a direction along the ridgeline **56a** of the recessed section **56** is produced by the load that is transmitted to the lower side section **48**. By such a load vector A, in the early phase of the collision, the lower side section **48** is deformed outward in the vehicle width direction to have a V-shape, and deformation of each of the side duct sections **44, 46** is thereby promoted.

(46) Next, a description will be made on an exemplary deformation state of the duct member **28** in the early phase of the collision according to this embodiment and operation of this embodiment in relation to this deformation with reference to FIG. 9 and FIG. 10. FIG. 9A and FIG. 9B are partial enlarged perspective views, each of which illustrates the exemplary deformation state of the duct member caused by the load input from the front of the vehicle to the vehicle having the vehicle front body structure according to this embodiment in the early phase of the collision. FIG. 10 is a partial enlarged perspective view for illustrating a deformation state of the side duct section at the time when the side duct section and the upper duct section in the duct member according to this embodiment are disengaged from each other.

(47) First, as illustrated in FIG. 9A and FIG. 9B, in the early phase of the collision, when the grille reinforcement **30**, which arranged on the inner side of the bumper fascia **8** and the front grille **10**, receives the collision load (indicated by a reference sign L in FIG. 9A and FIG. 9B) from the bumper fascia **8** and the front grille **10**, the load is transmitted to the side duct sections **44, 46** of the duct member **28** via the load transmission members **32** and the seal members **34**.

(48) In the early phase of the collision, the bumper reinforcement **20** also receives the collision load (the load L), and the collision load is transmitted to the side duct sections **44, 46** of the duct member **28**.

(49) A force that attempts to displace each of the side duct sections **44, 46**, which have received such a load, rearward in the vehicle with respect to the upper duct section **40** is applied thereto. First, as illustrated in FIG. 9A, such a force disengages the upper side section **50** in each of the side duct sections **44, 46** from the upper duct section **40** in the engagement section **54**. At this time, since a rear end of the upper side section **50** is supported by the radiator shroud **26**, as illustrated in FIG. 9B, the upper side section **50** is deformed in the vehicle width direction with the rear end thereof being a fulcrum.

(50) Meanwhile, in the early phase of the collision, the lower side section **48** is deformed in a manner to be bent outward in the vehicle width direction to have a V-shape due to the effect of the ridgeline **56a** of the above-described recessed section **56**. Along with such deformation of the lower side section **48**, a force that attempts to displace the upper side section **50** downward is applied to the upper side section **50**, which is arranged on top of the lower side section **48**.

(51) Here, as described above, in the early phase of the collision, the upper side section **50** is disengaged in the engagement section **54**. Thus, the bending deformation of the upper side section **50** is promoted.

(52) In this embodiment, strength of the duct member **28** itself against the collision load is reduced, so as to promote the deformation of the side duct sections **44, 46** as described above in the early phase of the frontal collision.

(53) Furthermore, in this embodiment, the duct member **28** is suppressed from receiving the large collision load by reducing the strength of the duct member **28**, just as described. In this way, the

deformation of the crash can **18** as the body-side shock absorbing member, which should be originally deformed to absorb the shock, is suppressed from being hindered.

(54) Moreover, in this embodiment, the frame sections **48a**, **50a** of the side duct sections **44**, **46** are made of the resin, and the inner surfaces **48b**, **50b** thereof are made of the rubber. Thus, the inner surfaces **48b**, **50b** do not hinder the deformation of the frame sections **48a**, **50a**, respectively.

(55) In addition, since the inner surfaces **48b**, **50b** of the side duct sections **44**, **46** are made of the rubber, the duct member **28** is less likely to receive the collision load in the early phase of the collision, which hinders the transmission of the collision load to the radiator **22** and the radiator shroud **26**. In this way, the deformation of the crash cans **18** is suppressed from being hindered.

(56) Next, a description will be made on effects of the front body structure of the vehicle **1** according to the embodiment of the present disclosure.

(57) First, the front body structure of the vehicle **1** according to the embodiment of the present disclosure includes the duct member **28** that surrounds the radiator **22** on the inner side in the vehicle width direction of the right and left pair of the crash cans **18**. The duct member **28** is the member that completely surrounds the radiator **22** and includes the side duct sections (the vertical wall sections) **44**, **46** and the upper duct section (the transverse wall section) **40**. Each of the side duct sections **44**, **46** of the duct member **28** and the upper duct section **40** has the at least one engagement section **54** for the mutual engagement and connection. Each of the side duct sections **44**, **46** of the duct member **28** and the upper duct section **40** are connected such that, when the duct member **28** receives the specified load from the bumper members (the bumper reinforcement **20** and the grille reinforcement **30**) of the body during the frontal collision, in the at least one engagement section **54**, each of the side duct sections **44**, **46** and the upper duct section **40** are disengaged from each other.

(58) According to this embodiment as described above, since the duct member **28** completely surrounds the radiator **22**, it is possible to ensure the air flow toward the radiator **22** as the heat exchanger. In addition, when the collision load is input from the front, in the two right and left engagement sections **54**, the side duct sections (the vertical wall sections) **44**, **46** and the upper duct section (the transverse wall section) **40** in the duct member **28** are disengaged from each other. Thus, it is possible to reduce the strength of the duct member **28** against the load from the front of the vehicle. In this way, the deformation of the crash cans **18** can be suppressed from being hindered during the frontal collision. As a result, according to this embodiment, it is possible to simultaneously improve cooling performance by the heat exchanger and exert collision performance of the vehicle.

(59) According to this embodiment, in each of the two right and left engagement sections **54**, the engagement vertical wall portion **50c** as a part of respective one of the side duct sections **44**, **46** is engaged with the notch **52** of the upper duct section **40** that is opened rearward. Thus, when the collision load is input to each of the side duct sections **44**, **46** from the front, respective one of the side duct sections **44**, **46** can be displaced rearward in the vehicle relative to the upper duct section **40** and can be disengaged from the notch **52** of the upper duct section **40**. In this way, it is possible to further effectively reduce the strength of the duct member **28** against the load from the front of the vehicle.

(60) According to this embodiment, the side duct sections **44**, **46** of the duct member **28** respectively have the frame sections **48a**, **50a** and the inner surface sections **48b**, **50b** with the lower strength than these frame sections **48a**, **50a**. The frame section **50a** of each of the side duct sections **44**, **46** in the two engagement sections **54** is partially connected to the upper duct section **40** in a manner capable of being disengaged therefrom. Thus, when each of the side duct sections **44**, **46** receives the collision load from the front, respective one of the frame sections **48a**, **50a** as the part of the side of the duct member **28** is deformed. In this way, it is possible to effectively reduce the strength of the duct member **28** against the load from the front of the vehicle.

(61) According to this embodiment, the recessed section **56** is formed with the ridgeline **56a** in the

manner to extend outward in the vehicle width direction. Thus, when each of the side duct sections **44, 46** receives the collision load from the front during the frontal collision, such a load is transmitted outward in the vehicle width direction along the ridgeline **56a** of the recessed section **56**. In this way, each of the side duct sections **44, 46** can be deformed outward in the vehicle width direction to have the V-shape. Along with such deformation, each of the side duct sections **44, 46** and the upper duct section **40** in respective one of the two engagement sections **54** can further effectively be disengaged from each other.

(62) According to this embodiment, the recessed section **56** of each of the side duct sections **44, 46** in the duct member **28** is formed at substantially the same height position as the bumper reinforcement **20** in the vehicle vertical direction of respective one of the side duct sections **44, 46**. Thus, it is possible to effectively transmit the collision load, which is input from the bumper reinforcement **20**, to the recessed section **56** of each of the side duct sections **44, 46**. In this way, each of the side duct sections **44, 46** can further reliably be deformed outward in the vehicle width direction to have the V-shape.

DESCRIPTION OF REFERENCE SIGNS AND NUMERALS

(63) **1** Vehicle **8** Bumper fascia **10** Front grille **18** Crash can **20** Bumper reinforcement **22** Radiator **26** Radiator shroud **28** Duct member **30** Grill reinforcement **32** Load transmission member **34** Seal member **40** Upper duct section **42** Lower duct section **44** Left side duct section **46** Right side duct section **48** Lower side section of the side duct section **48a** Frame section **48b** Inner surface section, rubber-made plane member **50** Upper side section of side duct section **50a** Frame section **50b** Inner surface section, rubber-made plane member **50c** Engagement vertical wall portion **52** Notch **54** Engagement section **56** Recessed section **56a** Ridgeline of recessed section

Claims

1. A vehicle front body structure comprising: a right and left pair of crash cans, each of which extends in a vehicle longitudinal direction; a radiator that is provided on an inner side in a vehicle width direction of the right and the left pair of crash cans; and a duct that surrounds the radiator on the inner side in the vehicle width direction of the right and left pair of crash cans, wherein the duct completely surrounds the radiator and has a vertical wall section and a transverse wall section, the vertical wall section and the transverse wall section of the duct have at least one engagement section in which the vertical wall section and the transverse wall section are engaged with and connected to each other, and the vertical wall section and the transverse wall section of the duct are connected such that, when the duct receives a specified load from a bumper of a body during a frontal collision, the vertical wall section and the transverse wall section are disengaged from each other in the at least one engagement section, wherein the transverse wall section of the duct is formed with at least one notch that is opened rearward, and the at least one engagement section is configured by engaging a part of the vertical wall section with the notch of the transverse wall section.

2. The vehicle front body structure according to claim 1, wherein the vertical wall section of the duct includes: a frame section; and an inner surface section having lower strength than the frame section, and a part of the frame section of the vertical wall section is connected to the transverse wall section in a manner to be disengaged therefrom in the at least one engagement section.

3. The vehicle front body structure according to claim 2, wherein the vertical wall section of the duct is formed with a recessed section that is formed with a ridgeline in a manner to extend outward in the vehicle width direction.

4. The vehicle front body structure according to claim 3, wherein the bumper of the body includes a bumper reinforcement that extends in the vehicle width direction on a front surface of a vehicle, and in a vehicle vertical direction of the vertical wall section, the recessed section of the vertical wall section of the duct is formed at substantially the same height position as the bumper

reinforcement.

5. The vehicle front body structure according to claim 1, wherein the vertical wall section of the duct includes: a frame section; and an inner surface section having lower strength than the frame section, and a part of the frame section of the vertical wall section is connected to the transverse wall section in a manner to be disengaged therefrom in the at least one engagement section.

6. The vehicle front body structure according to claim 1, wherein the vertical wall section of the duct is formed with a recessed section that is formed with a ridgeline in a manner to extend outward in the vehicle width direction.

7. A vehicle front body structure comprising: a right and left pair of crash cans, each of which extends in a vehicle longitudinal direction; a radiator that is provided on an inner side in a vehicle width direction of the right and the left pair of crash cans; and a duct that surrounds the radiator on the inner side in the vehicle width direction of the right and left pair of crash cans, wherein the duct completely surrounds the radiator and has a vertical wall section and a transverse wall section, the vertical wall section and the transverse wall section of the duct have at least one engagement section in which the vertical wall section and the transverse wall section are engaged with and connected to each other, and the vertical wall section and the transverse wall section of the duct are connected such that, when the duct receives a specified load from a bumper of a body during a frontal collision, the vertical wall section and the transverse wall section are disengaged from each other in the at least one engagement section, wherein the vertical wall section of the duct includes: a frame section; and an inner surface section having lower strength than the frame section, and a part of the frame section of the vertical wall section is connected to the transverse wall section in a manner to be disengaged therefrom in the at least one engagement section.

8. A vehicle front body structure comprising: a right and left pair of crash cans, each of which extends in a vehicle longitudinal direction; a radiator that is provided on an inner side in a vehicle width direction of the right and the left pair of crash cans; and a duct that surrounds the radiator on the inner side in the vehicle width direction of the right and left pair of crash cans, wherein the duct completely surrounds the radiator and has a vertical wall section and a transverse wall section, the vertical wall section and the transverse wall section of the duct have at least one engagement section in which the vertical wall section and the transverse wall section are engaged with and connected to each other, and the vertical wall section and the transverse wall section of the duct are connected such that, when the duct receives a specified load from a bumper of a body during a frontal collision, the vertical wall section and the transverse wall section are disengaged from each other in the at least one engagement section, wherein the vertical wall section of the duct is formed with a recessed section that is formed with a ridgeline in a manner to extend outward in the vehicle width direction.
